

RESEARCH ARTICLE

Assessment of Yield and Yield Component of Soybean Genotypes (*Glycine Max* L.) in North of Khuzestan

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Abstract

18 soybean genotypes were examined to investigate the relationships between some principal attributions of morphology with seed yield per soybean, by Random Complete Block Design (RCBD) study. This study was also carried out three replicates to gain reliable results. The results of variance analysis indicated that, there were significance differences among all soybean genotypes. Moreover, the results of correlated analysis revealed that biological yield (0.96), harvest index (0.92), and number of branches (0.92) had the uttermost correlation with seed yield. To data factor analysis, four independent variables justified 99 percent of all data. The first variable, seed yield, justified 96 percent of entire variance. Multiple-Regression Model with method Analytical Regression Model (step-by-step) was utilized to examine soybean seed yield. This model proved that biological yield, thousand seed weight, and harvest index entered into model respectively and justified 98.85 percent of variation of seed yield. Correlated coefficients of considered attributions were equal to 0.96, 0.78, and 0.92, respectively. All of these indexes had significant at 1% in statistical process. Therefore, these traits can be notability used in soybean breeding programs. Also, accordance to cluster analysis, the sample was divided into three groups.

Key words : Soybean, morphological traits, factor analysis, step-by-step regression

Introduction

Soybean, as strategic plant, can cope with nutria demands by the production of 40% protein and 20% oil (Monthly oil industry 2004). In Iran, approximately 100000 hectare of farmland has been planting in soybean under proper weather conditions. Moreover, in several provinces e.g., Golestan, Gilan, Mazandaran, and Ardebil around 2.2 tones soybean a hectare has been cultivated (Hymowitz and Kaizuma 2008). Therewith, Soybean as well as five oily plants (oil palm, rapeseed, cotton seed, peanut, and sunshine seed) can produce by 84% oil of the world (Top Fer et al. 1995). Hence, soybean and its attributions have substantial roles in economics. Hence, the cognition of attributions' relations and their interactions are crucial for all repairing plans (Acquah et al. 1992). It's also worthwhile to utter that soybean has a considerable interaction with daylight; therefor, exploring

convenient genotypes, determining appropriate period of cultivation, and varieties of soybean are essential factors to plant this seed. Two principals have been heeded to produce high qualified soybean, namely, variety of soybean with high potential genotype and variety of soybean with high adaptability.

Kumudini et al. (2002) compered the new and old variety of soybean, the results indicated that new cultivar of soybean had high quality due to the long durability of leaf at pods filling level and the escalation of dry materials at this level. The results of several studies also demonstrated that soybean with high-level of yield is reachable through high harvest index and the allocation of most of the photosynthetic materials into reproductive organs, whereas increasing surface of leaf until graining has contradictory relation with seed yield (Kumudini et al. 2002). In this vein, Jian Jin et al. (2010) studied 41 varieties of soybean and they found out that the duration (from sheathing to graining) was overriding to

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produce outstanding qualified soybean. Khan and Hatam (2000) illustrated that most of morphological attributions had meaningful and positive correlation with seed yield. Masudi et al. (2009) also reported that bush weight, numbers of seed and in bush had higher correlation with soybean yield. On the contrary, in study of Bangar et al. (2003) it was found that soybean yield had significance relation with weight of 100 grains, numbers of days from germination to 50% flowering, and time of cultivation. Henrico et al. (2004) as well as Akhtar and Sneller (1996) studies indicated that numbers of seed had meaningful correlation with seed yield per plant, whereas, this attribution had the highest direct impact on yield. Rezaizad (1999) investigated the existence relations between seed yield and its components and he explored that number of seed per plant, biological yield, and numbers of pod per plant had the most correlation with seed yield.

In this vein, the exact results cannot be reported through simple correlated coefficients due to the complicated relations among attributions. For this purpose, multi-variables statistical model is utilized to recognize the relations among attributions. Thus, data factor analysis as statistical method which revealed high correlations among variables is required to decrease data and get the fruits of data (Moqadam et al. 2004). The study on 14 attributions of 20 cultivar of soybean demonstrated four results through variables analysis method: first variable justified by 38.83 percent of data and it was called as natal variable; second variable justified by 21.4 percent of data and it was called as seed specifications variable; third variable justified by 17.35 percent of data and it was called as yield variable and the final variable justified by 7.5 percent of data and it was called as number of seed per pod variable (Sabokdast nodemi et al. 2010). Zhao et al. (1991) employed data factor analysis method in 12 important agricultural attributions by 16 soybean genotype in China. These attributions were classified into four groups. The first variable was contained number of seed per plant and numbers of pod per plant. The second variable was consisted plant height, number of node, height of the first pod from land and day numbers which have been needed to flower. The third variable was included number of pod per plant, hundred seed weight, and weight of seed per plant. The fourth variable was comprised number of branches. Motivated by previous research, it was resulted that cultivar of soybean had the significance impact on soybean seed yield. It is also notify to utter that, various sorts released

different yields accordance to environment conditions and their adaptabilities to those conditions. Thus recent studies tried to gain desired sort through modifying agricultural variables including history of planting, model of planting, etc. The current study also attempted to assess yield and yield component of prevalent soybean cultivars and employ these cultivars in future repairing plans.

Materials and Method

This study was cried out in experimental field of martyr Behesht Company in Dezful city (capital of Khuzestan Province in south west of Iran). Treflan spray was used (2.5 liters for one hectare) to battle against weeds. 200 kg/ha of potassium sulfate, 150 kg/ha Triple superphosphate and 50 kg/ha nitrogen fertilizer were used. The demanding nitrogen was amounted 150 kg/ha at fourth and fifth leaf levels and it was amounted 100 kg/ha at graining level to the plant, due to the lack of activated bacteria fixed soybean nitrogen; This RCBD study employed 18 genotypes of soybean and carried out three replicates. Each Crete contained 4 rows- 4 m in length and 60 cm in width- and the given gap between bushes was 5 cm. After complete growth, 10 bushes were chosen randomly from each Crete. The considered attributions consisting number of branches, number of pod per plant, plant height, pod length, number of node, thousand seed weight, biological yield, seed yield, and harvest index. All data was obtained through three-time assessment of attributions of selected bushes. This data was grouping through SAS 9.1 software (variance analysis) as well as Duncan Model (to compere average of data). Step-by-step Regression and SAS 9.1 software were used to analyze variables and, SPSS 18 software was utilized to analyze correlation and cluster.

Results and Discussion

The results of variance analysis (Table 1) proved that the impact of block and traits was significant for all attributions at 1 percent probable level. The most coefficient of genotype variation was belonged to number of node and the least coefficient was owned to biological yield. The results of

Table 1. Analysis of variance of measured traits in soybean cultivars.

S.O.V	df	Mean Square								
		Number of branches	Number of pod per plant	Plant height	Pod length	Number of node	Thousand seed weight	Biological yield	Seed yield	Harvest index
Block	2	3.02**	81.50**	59.46**	0.68**	4.24**	201.18**	523.01**	252.90**	2.22**
Treat	17	45.55**	3370.83**	2029.54**	15.30**	36.10**	22244.30**	181935.30**	27634.80**	713.46**
Error	34	0.09	2.04	2.60	0.20	0.14	4.47	19.84	11.45	0.28
C.V	-	5.25	2.82	4.64	3.77	6.64	1.39	1.08	2.32	1.94

** : significant at 1%

Table 2. Comparison of the studied traits in soybean cultivars.

Genotypes	Traits								
	Number of branches	Number of pod per plant	Plant height	Pod length	Number of node	Thousand seed weight	Biological yield	Seed yield	Harvest index
Sari	0.00h	0.00k	0.00k	0.00g	0.00g	0.00i	0.00j	0.00i	0.00h
Sahar	10.30b	92.30b	28.30i	4.00e	8.30b	171.30g	558.00d	206.70c	37.00cd
Gorgan3	10.30b	81.00c	37.00gh	5.50b	7.70c	240.70a	660.00c	204.70c	31.00g
Saman	8.30c	75.30d	101.30a	5.70b	8.30b	180.70f	408.30i	152.30fg	37.30c
Katol	7.70d	72.70e	62.70b	5.70b	9.00b	190.00e	455.00h	153.00f	33.70e
Kaspian	5.30f	56.00h	41.70ef	5.70b	5.30f	210.00d	456.00h	153.00f	33.60e
Nekador	6.30e	60.70g	39.30fg	5.70b	6.30de	210.00d	562.00d	173.30d	30.80g
2002	7.30d	65.30f	44.70cd	4.30d	6.30de	221.30c	505.00f	156.00f	30.80g
JK	4.70g	25.00j	23.00j	4.00e	5.30f	231.30b	482.00g	154.00f	31.90f
BP	4.30g	25.30j	23.00j	3.60f	6.30de	190.00e	455.00h	147.00g	32.30f
SG5	8.30c	70.70e	42.70de	6.70a	7.00d	180.00f	707.70b	305.00b	43.10b
HT	0.00h	0.00k	0.00k	0.00k	0.00g	0.00i	0.00j	0.00i	0.00h
SG10	6.30e	52.00i	36.30h	3.30f	9.00b	190.00e	406.30i	140.00h	34.40e
504	6.70e	65.70f	35.30h	5.10c	5.70ef	160.00h	460.00h	166.30e	36.30d
Salend	13.30a	101.00a	62.30b	4.60d	11.30a	180.70f	765.00a	337.00a	44.00a
SG2	0.00h	0.00k	0.00k	0.00g	0.00g	0.00i	0.00j	0.00i	0.00h
SG4	7.30d	67.30f	46.70c	4.60d	6.30de	179.30f	525.00e	176.70d	33.60e
PE	0.00h	0.00k	0.00k	0.00g	0.00g	0.00i	0.00j	0.00i	0.00h

Any two means that didn't share a common letter differ significantly from each other at 1% probability level.

comparing average of attribution in Table 2 revealed that the most number of branches (13.30), number of pod per plant (101.00), number of node (11.30), biological yield (765.00 kg/ha), seed yield (337.00 kg/ha) and harvest index (44.00 kg/ha) were existence in salend cultivar. The saman cultivar showed the most plant height (101.30) from farmland. On the other hand, the most pod length (6.70 cm) was observed in SG5 cultivar. The most thousand seed weight (240.70 gr) also belonged to Gorgan 3. Olser and Cartter (2004) stated that some components of yield which are consisting seed size, number of seed per pod and numbers of pod per plant, etc. are crucial factors in progression of soybean yield; therefore, genotypes which have been empowered with these high qualified constituents have much more potatoes genetically. Farahani Pad et al. (2012) demonstrated that the impact of cultivar was meaningful on thousand seed weight and seed yield in four cultivars. In most of products, yield is defined as the mixture of huge numbers of biological processes occurred during growing. Accordance to Ghorban zade neghab et al. (2013) study, Zane cultivar with 14.8 gr had the most weight of hundred seeds and Sahar cultivar with 9.2 had the least weight of hundred seeds among studied cultivars. The results illustrated that the Zane seed had the most weight of hundred seeds due to few numbers of seed per plant and lack of competition among seeds.

Correlation analysis

Determination of correlation analysis was one of the indexes to assess the existence relations among attributions. The result revealed that seed yield had the most correlation with biological yield (0.96) (Table 3). Correspondently, Masudi et al. (2009) Yunesi hamze khanlu et al. (2010), Namdari and Mahmudi (2013), as well as Iqbal et al. (2003) reported the meaningful correlation between seed yield and these four attributions: numbers of pod per plant, number of seed per pod, harvest index and number of branches. Similar findings were reached by Pedersen and Lauer (2004), Shibels et al. (1996) and Kumudini et al. (2001). Respecting the plant height, number of node onto cardinal branch, numbers of branches, number of pod per plant and weight of thousand seeds were effective factors on improvement of soybean yield; therefore, genotypes had more potential with these high qualified attributions. This lends evidence to previous studies which suggested that cultivar of Selend, SG5, and Gorgan 3 are superior proceed than others (Amaranthath et al. 1990; Das et al. 1989; Pendy et al. 1973; Rajput et al. 1986).

Factor analysis

The considerable studies were conducted to assess the impact of relations on effect of trait via analyzing coefficients to factor analysis. The recent research concentrated on causal analysis and determination of crucial criteria for repairing

Table 3. Simple phenotypic correlation coefficients among studied traits in soybean cultivars.

traits	Number of branches	Number of pod per plant	Plant height	Pod length	Number of node	Thousand seed weight	Biological yield	Seed yield	Harvest index
Number of branches	1								
Number of pod per plant	0.97**	1							
Plant height	0.75**	0.78**	1						
Pod length	0.85**	0.88**	0.66**	1					
Number of node	0.95**	0.92**	0.82**	0.84**	1				
Thousand seed weight	0.78**	0.76**	0.67**	0.84**	0.84**	1			
Biological yield	0.92**	0.88**	0.66**	0.87**	0.91**	0.89**	1		
Seed yield	0.92**	0.87**	0.65**	0.81**	0.90**	0.78**	0.96**	1	
Harvest index	0.89**	0.87**	0.76**	0.86**	0.94**	0.90**	0.94**	0.92**	1

** : significant at 1%

Table 4. Factor analysis by principal component method with varimax for the studied traits.

Trait	Rotated Factor Pattern			
	First factor	Second factor	Third factor	Fourth factor
NB	0.80	0.36	0.29	0.36
NP	0.66	0.56	0.25	0.41
PH	0.58	0.55	0.46	0.29
PL	0.52	0.45	0.51	0.43
NN	0.34	0.83	0.32	0.26
TSW	0.24	0.28	0.87	0.27
BY	0.47	0.31	0.47	0.66
SY	0.56	0.34	0.41	0.59
HI	0.39	0.54	0.31	0.56
Eigenvalue	76710.31	1867.71	513.11	169.13
Proportion	0.96	0.02	0.01	0.00
Cumulative	0.96	0.98	0.99	0.99

NB: Number of branches, NP: Number of pod per plant, PH: Plant height, PL: Pod length, NN: Number of node, TSW: Thousand seed weight, BY: Biological yield, SY: Seed yield, HI: Harvest index

soybean yield. In the current study, the results of analyzing 9 morphological attributions through cardinal factors highlighted four principal variables (Table 4). These four variables explained by 96%, 2%, 1%, and 0.21% of data diversities respectively and as whole, they clarified by 99% of data diversities. There is also a direct relationship between variables variance and variables value in data interpretations. In this vein, subscription rate was a part of variance variable which has been related to common variables. In addition, there was direct relationship between subscription rate and accurate rate (Henrico et al. 2004). By observing of revolved variable coefficients, it was found that the first variable coefficients, proceed variable, covered most of data and contained the big and positive coefficients of seed proceed, biological proceed, removal index, and tie numbers (Table 4). Similarly Yunesi hamze khanlu et al. (2010) examined variable analysis of 9

attributions within 33 mutated soybean lines. They illustrated that numbers of pod per plant, numbers of seed per plant and harvest index were crucial attributions to improve soybean yield. Moreover, Narjesi et al. (2008) tested 17 attributions of 30 soybean genotypes. The result proved that two variables of phenology and yield justified by 28.21% and 16.56% of data diversities, respectively. They also declared that harvest index and seed numbers had the biggest effect on soybean seed yield.

The second variable, yield component, contained the big and positive coefficients of biological yield, harvest index, thousand seed weight as well as pod length and also covered by 2% of data diversities.

The third variable contained the big and positive coefficients of number of node as well as plant height and covered by 1% of data diversities. It also contradicts with Kohkan and et al.

(2010) study in which 12 traits of 141 soybean line were examined. Based on their results, the first variable, phenological variable, consisted traits including yield, numbers of branches per plant, numbers of pod per plant, numbers of seed per plant, as well as seed weight per plant and covered by 29.18% of data diversities.

The fourth variable, pod variable, was contained the big and positive coefficients of attributions including numbers of pod per plant, numbers of branches as well as pod length and also covered by 0.21% of data diversities. In the same vein, Yahueian et al. (2010), found four main variables via factor analysis in stress conditions. These variables justified by 78.38% of data diversities. The first variable, phenologic-morphological variable, was covered most of the obtained data. The second variable or yield and yield component, the third variable or quality of seed, and the fourth variable were identified in stress conditions seed size.

In this study, step-by-step regression model was utilized. In this model after entering the new variable into the model, the old one was assessed by the model too. Hence, in this model, the most meaningful variable remained in functions. Furthermore, few variables but important ones were examined (Henrico et al. 2004) in this model. The results indicated that some attributions which are consisting biological yield, thousand seed weight, and harvest index entered into the model and covered by 98.85% of seed yield diversities

(Table 5). The inclination regression line also revealed that attributions of biological yield, thousand seed weight, and harvest index were 1 percent meaningful in statistical process. Some research declared that removal index is the best variable to justify soybean seed yield (Shukla et al. 1980; Weilenmann detau and Lugez 2000; Narjesi et al. 2008).

Results of cluster analysis (hierarchical grouping)

Accordance to grouping analysis, n people can form g groups ($g < n$). In other words, grouping analysis of genotypes must be classified based on similarity rate of separated groups (Zareh 2011). To select the cutting location and to determine optimal group numbers (g), the minimize variance method was used with below formula:

$$\text{Group numbers} = g = \sqrt{n/2} = \sqrt{18/2} = 3$$

Respecting to the results of Cluster analysis, cultivars of soybean were classified into three groups including Salend, SG5 and Gorgan 3 which had the best yield (Fig. 1). Alipour Yamchi et al. (2011) also examined genetic varieties and grouping of pea genotypes (*Cicer arietinum*). They stated that cluster analysis accordance to morphological attributions formed four independent groups of genotypes. Safari et al. (2008) formed three independent groups of peanut cultivars (*Arachis hypogaea*) via cluster analysis.

Table 5. Step-by-step regression analysis for seed yield as a function variable and other traits as independent variable.

step	trait	a	b ₁	b ₂	b ₃	Model R-square	C(p)
1	Biological yield	-9.41 ^{n.s}	0.37**			93.82	51.62
2	Thousand seed weight	2.13 ^{n.s}	-0.50**	0.53**		97.81	11.21
3	Harvest index	-1.37 ^{n.s}	-0.61**	0.44**	2.10**	98.85	2.21

a: the intercept; b: the slope; ^{n.s}: non-significant; **:significant at 1 %

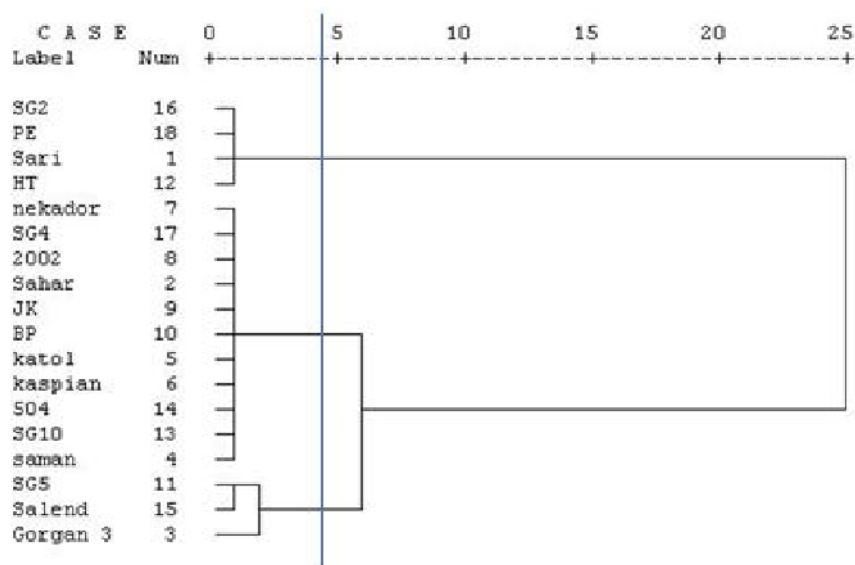


Fig. 1. Separate cluster of soybean varieties.

Conclusion

As result, analysis variable revealed that, there are several possibilities to repair soybean seed yield in future plans through combined selection of attributions.

Author contribution statement

SGH Conceived and designed research, wrote manuscript and acted as corresponding author. BAF and AN Supervised development of work, analyzed the data, helped in data interpretation and manuscript evaluation. NMN Conducted experiments, contributed new reagents and drafted the manuscript. All authors read and approved the final manuscript.

Compliance with ethical standards

Conflict of interest

The authors declare that they have no conflict of interest.

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